U1

miércoles, 24 de septiembre de 2025 10:07 a. m.

U1S1 Basic Concepts

sábado, 20 de septiembre de 2025
09:51 a. m.

What is machine learning?

Supervised
- Clasificacition Diagnostics, Costumer Retention, Fraud Detection, Image Classification
- Regression : Forecast, Predictions. Ex: Population Growth, life Expect. Weather, Market.

Unsupervised
- Dimensionality Reduction: Big Data Visualization Meaningfull Comprehension Feature Elicitation Structure Discovery
- Clustering Costumer Segmentation Target Marketing Recommender System

Reinforced.
- Real time Decisions
- Robot Navigation
- Learning Tasks
- Game AI
- Skill Aquisition

CLUSTERING

organization and classification of different objects data points or observations into groups or clusters based on similarity or pattern

Applications:
- Image segmentation
- Market segmentation
- Anomaly detection

CLUSTERING TASKS. Basic Steps

1. Feature selection
   that maximize relevant info
2. Proximity meassure
   How similar / different are the feature vectors
3. Clustering Criterion
   May be a cost function or other rules.
4. Validation of Results.
   Verify correctness of the results.
5. Interpretation of the results
   Maybe not us: but an expert on the field

FUZZY-CLUSTERING

X into m clusters by m functions u_j

Where each of the functions
- Each belonging function goes 0 to 1
  $u_j : X \to [0,1] \quad j = 1, \ldots, m$
- All the functions over a point are 1
  $\sum_{j=1}^{m} u_j(x_i) = 1 \quad i = 1, \ldots, N$
- More than 1 cluster bc the sum of the membership functions of 1 cluster must be less than N
  $0 < \sum_{i=1}^{N} u_j(x_i) < N \quad j = 1, \ldots, m.$

m-CLUSTERING

- X data in 1-dim space
  $X = \{x_1, \ldots, x_n\}$

We define $\mathcal{P}$ as m-clustering if it is the partition of X into m sets (clusters) $C_1, \ldots, C_m$ With conditions
- i $C_i \neq \emptyset \quad i = 1, \ldots, m$ no vacíos
- ii $\bigcup_{i=1}^{m} C_i = X$ todos suman los dato
- iii $C_i \cap C_j = \emptyset$ no estan en 2 o más

HOW DOES THE AMOUNT OF CLUSTERS BEHAVE?

$S(N, m)$ number of possible clusters for N vectors

String Number

$$S(N,m) = \frac{1}{m!} \sum_{i=0}^{m} (-1)^{m-i} \binom{m}{i} i^N$$

it grows rapidly

NOTE:

The similarity between objects depends on the Clustering Criterion

$S(N,m) = \frac{1}{m!} \sum_{i=0}^{m} (-1)^{m-1} \binom{m}{i} i^N$

it grows rapidly

U1S2: Proximity Meassures

domingo, 21 de septiembre de 2025

05:49 p. m.

DISIMILARITY

A DM d on X is a function

$$d: X \times X \to \mathbb{R}$$

when X is data such that:

- There is a lower limit $\exists d_0 \in \mathbb{R}$:
  $-\infty < d_0 < d(x,y) < \infty \quad \forall x,y \in X$
  meaning there is a finite distance between any given points on the data.
- $d(x,x) = d_0$ it is the least disimilar from itself $\forall x \in X$
- $d(x,y) = d(y,x) \quad \forall x,y \in X$
  it is symmetric

Also must satisfy these 2 conditions.

lower limit towards itself

- $d(x,y) = d_0 \iff x = y$

Triangle disimilarity

- $d(x,z) \leq d(x,y) + d(y,z) \quad \forall$

A disimilarity meassure is basically a distance

OTHER DISTANCES:

MAHALANOBIS:

DIST

$$d(x,y) = \sqrt{(x-y)^T \Sigma^{-1} (x-y)}$$

Where Σ is the covariance matrix of X

Properties:

- Invariant: to any nonsingular linear transformation
- Tends to form hyper ellipsoidal clusters
- No need for scale or standarizing
- Computationally expensive

CANBERRA

DIST

$$d(x,y) = \begin{cases} 0 & \text{for } x = y = 0 \\ \sum_{i=1}^{l} \frac{|x_i - y_i|}{|x_i| - |y_i|} & \text{for } x \text{ or } y \neq 0 \end{cases}$$

Properties:

Very sentitive to small shifts

Usefull when those are important

COSINE SIMILARITY

$$S_{COSINE}(x,y) = \frac{x^T y}{\|x\| \, \|y\|}$$

Where: $\|z\| = \sqrt{\sum_{i=1}^{l} z_i^2}$

Similarity

A SM on X is defined as

- $$s: X \times X \to \mathbb{R}$$

such that: there is an upper bound $\exists s_0 \in \mathbb{R}$

$-\infty < s(x,y) \leq s_0 < \infty \quad \forall x,y \in X$

Meaning a point will be at most the upper bound and if not have a finite sm to the other point.

- $s(x,x) = s_0$ each point is most similar to itself aka the upper bound
- $s(x,y) = s(y,x)$ it has to be symmetric $\forall x,y \in X$

To also be a similatrity meassure it must also

$$s(x,y) = s_0 \iff x = y \quad \text{bound to itself}$$

And this shit that is wierder than the triangle inequality.

$$s(x,y)\, s(y,z) \leq [s(x,y) + s(y,z)]\, s(x,z)$$

$$\forall x,y,z \in X$$

DMs for real value vectors

The weighted l_p metric or Minkowski distance

$$d(x,y) = \left(\sum_{i=1}^{l} w_i |x_i - y_i|^p \right)^{\frac{1}{p}}$$

Where x_i, y_i are the ith coordinates of the vectors x and y are vectors within X.

With the characteristics:

Invariant: to any translation and rotation only for $p = 2$ (Euclidian dist)

Dominate: features with large values and variance dominate other features

MINKOWSKI SPECIAL CASES

MANHATTAN: City block or cab distance

$$d(x,y) = \sum_{i=1}^{l} |x_i - y_i|$$

EUCLIDEAN: l_2 norm

$$d(x,y) = \sqrt{\sum_{i=1}^{l} (x_i - y_i)^2}$$

L-MAX: CHEBYSHEV

$$d_\infty(x,y) = \max_{1 \leq i \leq l} |x_i - y_i|$$

Could be shown

$$d_\infty(x,y) \leq d_2(x,y) \leq d_1(x,y)$$

U1S2: Proximity Meassures pt2

domingo, 21 de septiembre de 2025
07:03 p. m.

What is a CONTINGENCY TABLE?

Let x be a vector whole coordinates belong to a finite set
$F = \{0, 1, \dots, k-1\}$ where k is a positive integer
So there are exactly k^l vectors $\in F^l$

One can imagine these vectors as vertices in a l dimensional grid.

Consider: $x, y \in F^l$ and let

$$A(x,y) = [a_{ij}] \quad i,j = 0, 1, \dots, -1$$

be a $k \times k$ matrix, where the element a_{ij} is the number of places where the first vector has the i symbol

EXAMPLE: if $l = 6$, $k = 3$ and
$x = (0, 1, 2, 1, 2, 1)^T$
$y = (1, 0, 2, 1, 0, 1)^T$

Then the contingency matrix is

$$A(x,y) = \begin{vmatrix} 0 & 1 & 0 \\ 1 & 2 & 0 \\ 1 & 0 & 1 \end{vmatrix}$$

also $\sum_{i=0}^{k-1} \sum_{j=0}^{k-1} a_{ij} = l$

$\boxed{\text{Pearson's Correlation}}$ $r_{pearson}(x,y) = \dfrac{x_d^T y_d}{\|x_d\| \|y_d\|}$

Where $x_d = (x_1 - \bar{x}, \dots, x_l - \bar{x})$ and
$\bar{x} = \frac{1}{l} \sum_{i=1}^{l} x_i$ i is the ith coordinate.

Values for pearsons correlation coeficient go from $[-1, 1]$
Unable to detect the magnitude of the diff's

$\boxed{\text{DM for discrete value vectors}}$

· HAMMING DISTANCE
the number of places where the vectors differ

Using X matrix $d_H(x,y)$ is

$$d_H(x,y) = \sum_{i=0}^{k-1} \sum_{j=0}^{k-1} a_{ij}$$

the summation of all the off diagonal elements of A which indicates where x and y differ
Only for same length vector strings.

Note on l_1 it is the same as hamming when the vectors are binary values.

TANIMOTO DISTANCE: Inspired in comparisson of sets if X and Y are two sets:

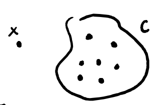
$$\frac{|X \cap Y|}{|X| + |Y| - |X \cap Y|} = \frac{|X \cap Y|}{|X \cup Y|}$$

the ratio to the number of elements each set has in common.

U1S3: Proximity Functions on Sets

domingo, 21 de septiembre de 2025
08:14 p. m.

Between <u>Point and a set</u>



In many algorithms to be able to add the point to a cluster means to see how far it is from the set.
$\phi(x, C)$
2 ways to do it
- All points in C contribute mean, min, max etc
- A representative oint
  medoid: representative point $\in$ set.
  centroid: mean point of the set.

<u>SET REPRESENTATIVE</u> on clusters that are circular or hyperspherical (computer vision, apps)

<u>SET REPRESENTATIVES</u>

When the points on a set are non-linear vectors. Such cases we use a hyperplanes as representatives

$$\sum_{j=1}^{l} a_j x_j + a_0 = a^T x + a_0 = 0$$

where $x = (x_1, \dots x_l)^T$, $a = (a_1, \dots a_l)^T$ weighted vector for H
In the case of euclidean distances between two points

$$d(x, H) = \frac{|a^T x + a_0|}{\|a\|}$$

and where $\|a\|$

Between <u>SUB-SETS</u>: Proximity for subsets of X

Let U be a subset of subsets of X. That means
$D_i \subset X$, $i = 1, \dots, k$ and
$U = \{D_1, \dots, D_k\}$
A proximity meassure ϕ on U is a fuction
$\phi : U \times U \to \mathbb{R}$
that satisfied
- $\exists d_0 \in \mathbb{R}$ $-\infty < d_0 \leq d(D_i, D_j) < \infty$ $\forall D_i, D_j \in U$
- $d(D_i, D_i) = d_0$ $\forall D_i \in U$ no dist to itself.
- $d(D_i, D_j) = d(D_j, D_i)$ symmetric

Between 2 sets

- Max proximity function
  max 2 points on the sets
- min proximity function

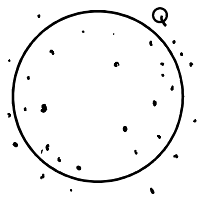
SET REPRESENTATIVE on clusters that are circular or hyperspherical (computer vision apps)

Q is a hypersphere.

$$(x-c)^T(x-c) = r^2$$

Where c is the center of the hypersphere & r is radious

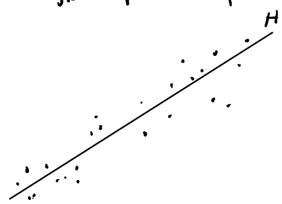
and $d(x, Q) = \min_{z \in Q} d(x, z)$



$$d(x, H) = \frac{|a^T x + a_0|}{\|a\|}$$

and where $\|a\|$

$$\|a\| = \sqrt{\sum_{j=1}^{l} a_j^2}$$



- max proximity function
  - max 2points on the sets
- min proximity function
  - same idea
- average proximity function
- mean
  - representative